

Egor Petrov

Machine Learning Researcher

petrov.egor.d@phystech.edu | [GitHub: modernTalker](#) | [Google Scholar](#)

Education

Moscow Institute of Physics and Technology (MIPT) *Moscow, 2022 – 2026 (Expected)*
B.S. in Applied Mathematics and Informatics
Data Science Track (MIPT & Yandex School of Data Analysis) *Moscow, 2024 – 2026*
Specialized curriculum in ML, DL, RL, CV, NLP, RecSys, Bayesian Statistics and Time Series Analysis.

Research & Engineering Experience

Yandex Research *Jul 2025 – Present*

ML Researcher, ML Research Residency, Moscow

- Spearheading research on asynchronous pipeline parallelism for efficient large-scale model training.
- Designing and analyzing novel algorithms for highly distributed systems to enhance scalability and performance.

Laboratory of Mathematical Methods for Optimization, MIPT *Feb 2024 – Present*

ML Researcher, Moscow

- Pioneered memory-efficient zeroth-order optimization methods for fine-tuning LLMs, achieving a **50% reduction in memory footprint**.
- Developed and theoretically analyzed novel stochastic and zeroth-order algorithms for decentralized optimization, validated on variety of domains.

Yandex *Aug 2024 – Jan 2025*

ML Engineer, Personalization Quality R&D Group, Moscow

- Engineered a production pipeline integrating Vowpal Wabbit-based online features into a large-scale CatBoost model, achieving a **0.1% uplift in core ranking metrics** (AUC, nDCG).
- Implemented a parallelized, MapReduce-style data converter that **accelerated data processing by 20x** and reduced model training time from hours to minutes.

Publications

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- **Leveraging Coordinate Momentum in SignSGD and Muon: Memory-Optimized Zero-Order LLM Fine-Tuning**
E. Petrov, G. Evseev, A. Antonov, A. Veprikov, P. Plyusnin, N. Bushkov, et al.
ICML Workshop on Tiny Titans, 2025. [Paper]
 - **Sign-SGD is the Golden Gate between Multi-Node to Single-Node Learning...**
D. Medyakov, S. Stanko, G. Molodtsov, P. Zmushko, G. Evseev, E. Petrov, et al.
Submitted to ICLR 2026 [Paper]
 - **When Extragradient Meets PAGE: Bridging Two Giants to Boost Variational Inequalities**
G. Molodtsov, V. Parfenov, E. Petrov, E. Grigoriy, D. Medyakov, A. Beznosikov.
Conference on Uncertainty in Artificial Intelligence (UAI), 2025. [Paper]
 - **Shuffling Heuristic in Variational Inequalities: Establishing New Convergence Guarantees**
D. Medyakov, G. Molodtsov, E. Grigoriy, E. Petrov, A. Beznosikov.
ICOMP 2024. Submitted to Q1 Journal JOTA. Awarded 1st place at Neuroinformatics 2024. [Paper]
 - **Zero Order Algorithm for Decentralized Optimization Problems**
A. Veprikov, E. Petrov, G. Evseev, A. Beznosikov.
AI Journey 2024. Published in Doklady Mathematics. [Paper]
 - **Sampling of Semi-Orthogonal Matrices for the Muon Algorithm**
E. D. Petrov, G. V. Evseev, A. V. Antonov, A. S. Veprikov, N. A. Bushkov, S. V. Moiseev, A. N. Beznosikov.
Submitted to Q2 Journal Doklady Mathematics.
 - **Closing the Curvature Gap: Full Transformer Hessians and Their Implications for Scaling Laws**
Egor Petrov, Nikita Kiselev, Vladislav Meshkov, Andrey Grabovoy.
Submitted to ICLR 2026. [Paper]

Projects & Leadership

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- **Lead Developer, ZO-Library** *[GitHub]*
Engineered a comprehensive open-source library for zeroth-order (ZO) optimization, implementing a wide range of state-of-the-art algorithms. Designed with a user-friendly, `torch.optim`-style API for seamless integration into existing ML pipelines.

- **Public Talks**

Gave public talks at scientific seminars and reading groups

- **Memory-efficient LLM training: Existing methods for effective training of large language models** [11.2025]

Scheduled a talk at the Open Reading Group Seminar of *Basic Research in Artificial Intelligence Laboratory* at MIPT.

- **SimCLR: overview and implications** [06.2024]

Gave a talk at *A.A. Kharkevich Institute of Information Transmission Problems of the Russian Academy of Sciences* as part of the Scientific Literature. Seminars

Teaching Experience

- **Central University** – Teaching Assistant [2025 – Present]

Serve as a teaching assistant for the *Introduction to Artificial Intelligence*, *Scientific Studio*, and *Python for Data Analysis*; responsibilities include mentoring scientific projects and checking homework assignments.

- **Moscow Institute of Physics and Technology** – Teaching Assistant [2024 – 2025]

Served as a teaching assistant for the *Algorithms and Data Structures* course; responsibilities included mentoring students' progress and providing consultations.

Technical Skills

- **Programming:** Python, C++, SQL
- **ML/DL Frameworks:** PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Transformers
- **Tools & Infrastructure:** Git, Docker, Apache Hadoop, Kafka, Airflow, Vowpal Wabbit, CMake
- **Languages:** Russian (Native), English (Professional Working Proficiency)